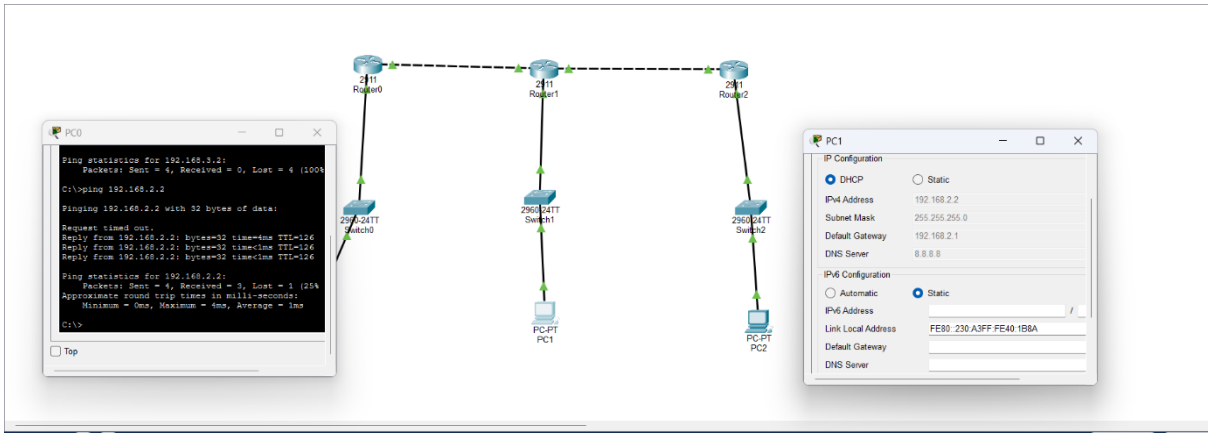
Day 1 – 5

1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.



Use this matching table for five PCs in the same subnet:

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.2.2	255.255.255.0	192.168.2.1
PC1	192.168.2.3	255.255.255.0	192.168.2.1
PC2	192.168.2.4	255.255.255.0	192.168.2.1
PC3	192.168.2.5	255.255.255.0	192.168.2.1
PC4	192.168.2.6	255.255.255.0	192.168.2.1

Steps:

1. Open each PC → Desktop → IP Configuration → Static.
2. Enter the corresponding IP address, subnet mask, and gateway from the table.
3. On PC0, open Command Prompt and run ping 192.168.2.3.
4. Repeat the ping command for the other PCs to verify connectivity.

Result: The PCs are assigned unique IP addresses within the same subnet. Successful ping replies confirm connectivity between them.

2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

DHCP Server Configuration in Cisco Packet Tracer

Procedure:

1. Connected three PCs to switches and configured the network using three routers in Cisco Packet Tracer.
2. Enabled DHCP on each PC by selecting Desktop → IP Configuration → DHCP.
3. The DHCP server automatically assigned IP addresses, subnet masks, default gateways, and DNS server addresses to the PCs.
4. Verified the assigned addresses in the IP Configuration windows.

IP Address Allocation Table

Device	Assigned IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.2.2	255.255.255.0	192.168.2.1
PC2	192.168.3.2	255.255.255.0	192.168.3.1

Result: All three PCs received unique IP addresses from their respective network ranges through DHCP, as shown in the attached screenshot.

3.Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

Procedure:

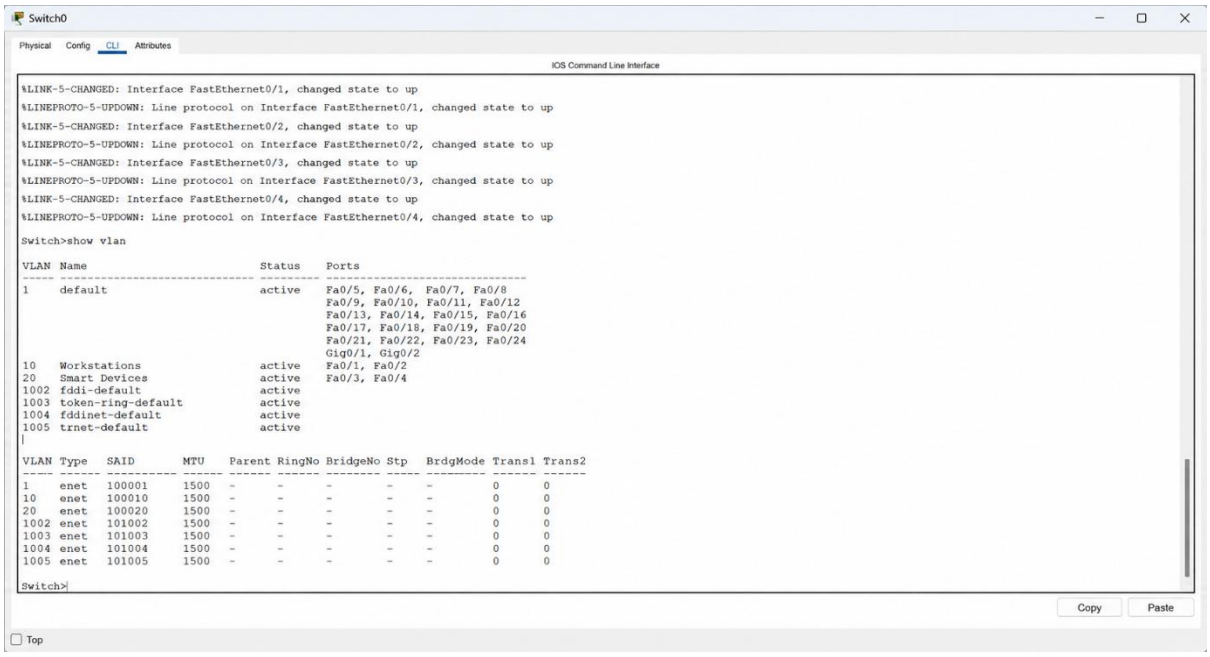
1. Configured a DNS server on the local network and enabled the DNS service.
2. Added a DNS record mapping the hostname musicplayer.local to the local web server's IP address 192.168.1.10.
3. Configured the connected devices to use the DNS server's IP address.
4. Hosted a web page on the server and ensured that the web service was enabled.
5. Opened a browser on a connected device and entered http://musicplayer.local.
6. Verified that the local web page opened successfully.

DNS Record Table

Hostname	IP Address	Record Type
musicplayer.local	192.168.1.10	A

Result: The custom hostname musicplayer.local resolved to the local web server's IP address, allowing connected devices to access the web page using the hostname instead of the IP address.

4. Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.



VLAN Table

VLAN ID	VLAN Name	Assigned Ports	Devices
10	Workstations	Fa0/1, Fa0/2	PC0, PC1
20	Smart Devices	Fa0/3, Fa0/4	PC2, PC3

Steps

1. Created VLAN 10 (Workstations) and VLAN 20 (Smart Devices) on the switch.
2. Assigned two devices to each VLAN using access ports.
3. Verified the VLAN configuration using the show vlan command.
4. Tested connectivity using ping between devices in the same VLAN and across different VLANs.

Result

Devices in the same VLAN can communicate with each other. Devices in different VLANs cannot communicate directly without inter-VLAN routing.

5. Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

NAT Configuration and Verification

Procedure:

1. Configured the router with an inside interface connected to the LAN and an outside interface connected to the external network.
2. Configured NAT overload (PAT) to allow multiple LAN devices to share one public IP address.
3. Marked the LAN interface as ip nat inside and the external interface as ip nat outside.
4. Configured an access control list to identify the private IP addresses allowed to use NAT.
5. Tested connectivity from a LAN PC using the ping command to reach an external server.
6. Verified the NAT translations on the router using show ip nat translations.

Result: NAT translates private IP addresses into the router's outside IP address, allowing multiple LAN devices to share a single public IP address when accessing external networks.